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Participatory watershed management programmes for sustainable development: Experiences from IWDP

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ABSTRACT: Participatory watershed management, enshrined in the new guidelines of Govt. of India calls upon people's participation in the planning, implementation and equitable sharing of benefits and responsibilities. It has been successfully demonstrated through the development of six watersheds in different agro-ecological regions of the country having diverse physiographic, climatic and socio-economic conditions. The watersheds were developed as per Integrated Wastelands Development Programme (IWDP) guidelines. The analysis of data has revealed that watershed interventions on arable and non-arable lands in a participatory mode significantly reduced runoff and soil loss which in turn improved crop productivity on an average by 28%. Various indicators employed to assess the impact of different activities showed significant improvement in biophysical, participatory and socio-economic attributes. Community organization had a catalytic effect on people's participation and contributions, which on an average occurred to the extent of 62% and 27% respectively. The average annual income per family increased by 49% through employment and income generating activities in the watersheds. In spite of drought conditions prevailing in some of the watersheds for few years, the projects were economically viable with overall BCR of more than 1.14. Thus, participatory watershed management following new guidelines is the right approach to maximize productivity on sustained basis through efficient use of natural resources and to achieve economic, livelihood and environmental security.

Key words: Degradation; Indicators; Participatory; Sustainability; Watershed management

Very few countries have experienced rapid economic growth without agricultural growth either preceding or accompanying it. But this growth is inevitably at the cost of environmental degradation. According to the estimates of actual land use and vegetation cover by the National Remote Sensing Agency and Forest Survey of India based on satellite imagery, 80 m ha out of 142 m ha area under cultivation is substantially degraded and about 40 m ha out of 75 m ha of forest land has a canopy cover of less than 40%. Nearly 11 m ha of pasture land is also substantially degraded. Thus a total of 131 m ha, representing about 40% of the country's landmass, has productivity well below its actual potential. The steady growth of human as well as livestock population, widespread incidence of poverty, and current phase of economic and trade liberalization are exerting heavy pressure on India's limited land resources to meet competing demands in forestry, agriculture, pasture, human settlements and industries. This has led to serious land degradation problems to an extent of about 187.8 m ha (57% approximately) in one way or the other (Sehgal and Abrol 1994).

Balancing of production and conservation, as inspired by Agenda 21 of the Earth Summit (UNCED 1992), at many spatial and temporal scales, requires active involvement of local stakeholders and communities in development programmes. This resulted in a paradigm shift from top down to bottom-up approach as the role of the Government reversed from the programme executioner to a facilitator, advisor and supporter. Moreover, it was realized that conservation technologies required investment by stakeholders, individually as well as collectively. To meet these ends, watershed is the accepted unit for planning and execution of various developmental programmes as it signifies a layered natural and social phenomenon (multi scale, diverse user, complex resource), which is also appreciated by laypersons, policy makers and conservationists. Evaluation of watershed management programmes in 10 major agro-ecological regions of India indicated that due to various interventions, soil loss decreased by 10-80%, runoff by 2-42%, while the productivity of arable lands increased by 4.2 to 15.4 q ha^{-1} (Ram Babu *et al.* 1997). These programmes were found to be economically viable with benefit:cost ratio varying from 1.10:1 to 2.94:1 with tremendous positive environmental externalities. Integrated development of watersheds helps in moderating the floods downstream, mitigating the effects of droughts and improving soil moisture regimes for biomass production besides augmenting ground water recharge (Samra 1997).

People's participation has now been recognized as an integral component of India's watershed management programmes, which have undergone a paradigm shift from externally driven and centrally controlled approach to people-centered approach based on the recommendations of the Hanumantha Rao Committee (GOI 1994). Based on these recommendations, Government of India issued new "Guidelines for Watershed Development" in respect of schemes, projects and programmes within its mandate. Empowerment of community, gender neutrality, equity, transparency and management of common property resources by village level institutions are the paradigms enshrined in

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the new Guidelines (Sharda 2002). These Guidelines, effective from 1st April 1995, envisaged a fixed work programme to be followed by District Rural Development Agencies/Zilla Parishads.

According to Wastelands Atlas of India 2000, the total wastelands area covered in 584 districts is 63.85 m ha (including 14 m ha of degraded forests) which accounts for 20.17% of the total geographical area of the country. The Department of Land Resources, Ministry of Rural Development (MoRD), Government of India under its Integrated Wastelands Development Programme (IWDP) identified Central Soil and Water Conservation Research and Training Institute, Dehradun as a Project Implementing Agency (PIA) during 1997 to develop six watersheds through its Regional Research Centres following new Guidelines and test their applicability under field conditions through people's institutions. The major objectives of the study were promotion of economic development of the village community which is directly or indirectly dependent on watershed through optimum utilization of natural resources, mitigation of adverse effects of calamities like drought, checking of land degradation, and generation of employment through development of human and other economic resources of the watershed and income generating activities. It was also envisaged to sustain community action after the withdrawal phase for the operation and maintenance of the assets created and further develop the natural resources through simple and affordable technological solutions by blending indigenous knowledge with modern techniques. Lastly, special emphasis was laid on the improvement of the economic and social conditions of the resource poor and disadvantaged sections of the watershed community such as asset-less, land-less and women through more equitable distribution of the benefits of land and water resources and greater access to income generating opportunities.

MATERIALS AND METHODS

Description of Watersheds

The six watersheds are located in diverse agro-ecological regions of the country, namely Eastern Ghats, Western Ghats (Nilgiris), Shiwaliks (Himalayan foot-hills), Bundelkhand region, Western Coast Gujarat Plain and Chambal Ravines (Fig. 1). The morphological characteristics and socio-economic features of the six watersheds are presented in Table 1 (Katiyar *et al.* 2004; Kumar *et al.* 2004; Patnaik *et al.* 2004; Sharma *et al.* 2004; Sikka *et al.* 2004; Singh *et al.* 2004). The elevation ranges from 35 m to 1329 m (amsl) indicating diverse geo-climatic settings of the

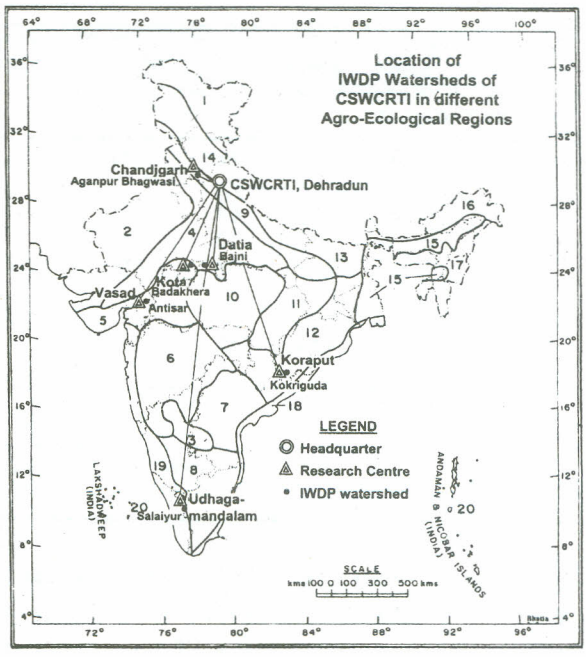


Fig. 1. *Location of IWDP watersheds alongwith Research Centres of Central Soil and Water Conservation Research and Training Institute, Dehradun in different Agro-Ecological Regions*

watersheds. Agriculture and labour account for 70 to 100 % of the stakeholders' occupation indicating high dependence of the watershed community on natural resources for their livelihood. Out of the six watersheds, four watersheds have average size of land holdings less than or equal to 2 ha signifying that majority of the farmers fall in small and marginal category, with average family income of Rs. 1,000/- to Rs. 2,500/- per month.

Present land use and natural resource degradation

More than half to a maximum of 86% of the area in the watersheds is under cultivation (Table 2) except in Bajni watershed, which is having only one-fourth of its total area under agriculture. Majority of Bajni watershed's area is under scrub forest and community land used for grazing cattle. In the five agriculture dominated watersheds, the cultivated land is mostly rainfed and assured irrigation was available in less than 20% of the area. In case of uncultivated area of the watersheds, there is a variation in land use from no forest to 31% area under forest/horticulture trees. A similar trend was observed in case of wastelands, which constituted 0 to 18% of the watershed areas. Runoff and soil loss in the watersheds were recorded by constructing gauging structures and employing standard procedures. Owing to various land-based activities in the watersheds and poor vegetative cover, 25 to 50% of the total rainfall was lost as surface runoff, except in Salaiyur watershed, where 4.5 to 7.2% runoff was reported. High runoff resulted in soil loss of 13 to 40 t ha^{-1} yr^{-1} in the watersheds.

Table 1. *Details of IWDP watersheds implemented by Research Centres of CSWCRTI, Dehradun*

Watershed	District & state	Agro-ecological region	Area (ha)	Elevation range (m) amsl	No. of families	Population	Major occupation (%)		Average holding size(ha)	Average annual income (Rs family^{-1})	Average milk production (lt family^{-1})
							Agri-culture	landless labour			
Aganpur-Bhagwasi	Patiala (Punjab)	9	550	80-286	166	1150	37	33	1.72	30,523	745
Antisar	Kheda (Gujarat)	5	812	25-35	500	2104	88	12	3.12	16,247	677
Badakhera	Bundi (Rajasthan)	5	682.5	150-173	117	1117	78	22	3.23	25,811	613
Bajni	Datia (Madhya Pradesh)	4	532	263-284	176	993	52	48	1.10	18,597	653
Kokriguda	Koraput (Orissa)	12	317.5	880-1329	78	249	93	7	2.15	12,155	12
Salaiyur	Coimbatore (Tamil Nadu)	19	513	370-472	314	1314	56	38	1.98	19,837	1146

Table 2. *Area under different land uses in the IWDP watersheds*

S. No.	Particulars	Aganpur-Bhagwasi	Antisar	Badakhera	Bajni	Kokriguda	Salaiyur
1.	Cultivated land						
	Rainfed farming	393.3 (71.5)	586.1 (72.2)	338.7 (49.6)	70.0 (13.2)	161.7 (50.9)	340.5 (66.4)
	Irrigated farming	80.0 (14.5)	112.5 (13.9)	40.2 (5.9)	60.0 (11.3)	6.0 (1.9)	65.6 (12.8)
	Sub-total	473.3 (86.1)	698.6 (86.0)	378.9 (55.5)	130.0 (24.4)	167.7 (52.8)	406.1 (79.2)
2.	Forest and horticultural land	20.6 (3.7)	NA	NA	166.0 (31.2)	140.6 (44.3)	Neg.
3.	Pasture land	2.0 (0.4)	NA	NA	169.0 (31.8)	Neg.	Neg.
4.	Wasteland	34.6 (6.3)	NA	NA	64.0 (12.0)	2.0 (0.6)	94.3 (18.4)
5.	Habitation, road etc.	19.5 (3.5)	NA	NA	3.0 (0.6)	7.2 (2.3)	12.6 (2.5)
	Sub-total	76.7 (13.9)	113.4 (14.0)	303.6 (44.5)	402.0 (75.6)	149.8 (47.2)	106.9 (20.8)
	Total	550.0 (100.0)	812.0 (100.0)	682.5 (100.0)	532.0 (100.0)	317.5 (100.0)	513.0 (100.0)

Note: Figures in parentheses are the percent of the total watershed area; NA - Not Available; Neg. - Negligible

Community Organization and Capacity Building

Community organization is an important aspect of any people centered development project. Therefore, the Watershed Development Team (WDT), a multi-disciplinary team of Institute's scientists, including lady scientists (wherever present) and social scientists conducted several rounds of meetings with watersheds' communities. Participatory Rural Appraisal (PRA) exercises were conducted to generate awareness about the project and its mode of execution. Further, exposure visits to successfully executed watersheds by the Research Centres were conducted to reinforce the confidence of the stakeholders and several Entry Point Activities (EPAs) were undertaken by the WDT in each watershed, with and without local contributions. Watershed level social organizations or local level people's institutions, well represented by various sub-communities of the watersheds, were constituted for decision making in all aspects of watershed development. These self-sustaining watershed level institutions were created to take over the activities after the withdrawal phase of the project.

Four types of groups were organized at watershed level. Self-Help Groups (SHG) of poor and landless people having a common identity and/or interest were formed in the watersheds (Table 3). Alternative income generating activities were undertaken for upliftment of the landless and resource poor farm families in each watershed through these groups. Seed money was distributed as a loan to SHGs from a Revolving Fund set up to provide initial financial help. The seed money was recovered and again distributed to other SHGs. SHGs formed ranged from 4 to 28 with 5 to 18 members in each group. User Groups (UG) were constituted of farmers owning land to take up different land based activities (Table 3). Land development activities like levelling and bunding, as well as construction of gabion and loose boulder structures for gully control were also undertaken. For water resource development, construction, desilting and deepening of percolation ponds, and management of underground pipeline system were executed. The number of User Groups varied from 2 to 6 in the watersheds. Watershed Association (WA), the apex General Body, comprising of watershed community members or stakeholders who agreed to participate in the watershed development programme met regularly to approve the strategic plan and review programme during implementation phase.

Watershed Committee (WC) consisting of nominated members from SHGs and UGs, in addition to one member each

Table 3. *Self help and user groups constituted in the IWDP watersheds*

Watershed	Self help groups	Total members	Self help groups' activities	User groups	User groups' activities
Aganpur-Bhagwasi	3	32	Tailoring & embroidery, carpet weaving, artisans	5	Land development, crop improvement, horticulture /agro-forestry development, gully control
Antisar	4	55	Grazing land protection, forest / home industries development, social welfare	5	Bunding and land levelling, water management, animal husbandry/agriculture/horticulture development
Badakhera	10	50	Mason, barber, blacksmith, carpenter, washerman, tailor, embroider, vegetable/milk vendor, grocer	6	Bunding & levelling, masonry /gabion & loose boulder structures, water resource / horticulture / crop development
Bajni	14	79	Business, selling of crop, vegetables, fruits & milk	4	Horticulture, dairy and fodder, vegetable/ crop production
Kokriguda	3	49	Mushroom, nursery raising & vegetable cultivation	2	Hillock development and management of underground pipe line system
Salaiyur	28	124	Coir making, coconut frond weaving, plastic bags, knitting, tailoring, vegetable/coconut/ cattle feed selling, sheep rearing, petty shop, thrift society, power sprayer, iron box, over lock/ fibre slicer machine	3	Construction of check dams and percolation ponds, desilting and deepening of percolation ponds, agroforestry

from Gram Panchayat and the WDT, carried out day to day activities of the watershed development programme subject to overall supervision and control of the WA. It also resolved conflicts arising out of the project activities. WC had 10 to 15 members with one elected President and had adequate representation of women and landless/SC/ST/OBC community (Table 4). A Watershed Secretary (WS) was also identified by the WC with the assistance of WDT for maintaining records related to physical and financial progress, disbursement of wages to the labourers, purchase of material, maintenance of the measurement book etc.

A joint Watershed Project Account was opened for each watershed in a nearby bank with the Chairman (WC), Nodal Scientist (WDT) and WS as co-signatories. This ensured transparency in the utilization of funds earmarked for various works as major payments were made through cheques except for labour wages and petty works. Transparency was further ensured by maintenance of accounts and records by the WS and WC members, for which they were imparted training. Trainings in tailoring, mushroom cultivation, agriculture, soil and water conservation etc. were also imparted to needy and desiring watershed inhabitants to upgrade their skills and improve their livelihoods. To ensure sustainability of the project activities after the withdrawal of the PIA, a Watershed Development (Corpus Fund) Account was opened as a fixed deposit/interest bearing joint account in a bank for depositing all the contributions received during the project. Contributions in cash were directly deposited whereas in case of those in the form of labour or kind were deposited by withdrawing equivalent amount from the Watershed Project Account. Rate of contributions by the watershed community varied depending upon the location and type of project activity. Fines or fees for services rendered were also deposited in this account.

Developmental Interventions

Developmental interventions were undertaken in the watersheds as per the problems, needs and priorities of the watershed community and their technical feasibility. A mix of engineering and biological measures was adopted for treatment of the watersheds to make it cost effective within the prescribed norms. The various development activities undertaken were :

Arable lands

Activities for development of arable lands included were mechanical measures like land shaping, levelling, terracing and bunding in all the watersheds except Salaiyur. Land levelling and terracing proved very effective and popular among the watersheds' farmers as well as in adjoining areas of watersheds. This activity was undertaken in 2 to 142 ha of

Table 4. *Watershed committees constituted in the IWDP watersheds*

Watershed	Representatives of				Total Members
	Panchayat	Women	Landless/ SC/ST/OBC	Others	
Aganpur-Bhagwasi	2 (1OBC man, 1 SC woman)	2 (SC)	3	8	15
Antisar	3 (1woman)	2	2	6	13
Badakhera	0	1 (SC)	6	3	10
Bajni	2 (1 OBC man, 1 ST woman)	0	8	1	11
Kokriguda	0	2 (ST)	9	1	12
Salaiyur	2 (2 OBC men)	3 (OBC)	7	1	13

arable land falling mostly under private ownership to check soil erosion and enhance productivity. Even in Salaiyur, land levelling was initially undertaken in 0.25 ha, but it was discontinued due to shallow soil depth.

Construction/strengthening of contour, graded and field bunds was another important mechanical activity undertaken for development of 8 to 30 ha of arable land in the watersheds. In Kokriguda watershed, to reduce the scouring velocity of the runoff, bunds were planted with grasses of sambuta (*Eulaliopsis binata*) and vetiver (*Vetiveria zizanioides*), hedges of Assam shade (*Indigofera zollingeriana*) and *Gliricidia*, and fruit trees (papaya and drum stick). Napier grass (*Pennisetum purpureum*) was planted on bunds in Aganpur-Bhagwasi watershed for stability and fodder production.

For safe disposal of excess runoff water from cultivated fields, disposal structures were constructed in Aganpur-Bhagwasi (45 structures) and Kokriguda (20 structures) watersheds. Drip irrigation was introduced in Salaiyur watershed for efficient management of precious water resource keeping in view the low irrigation efficiency, fast depleting groundwater resource and farmers' preference for intensive irrigation crops like coconut, banana and sugarcane. The system was also adopted for mulberry, mango and tamarind plantations.

Reclamation of alkali soils in 5.1 ha area was undertaken in Aganpur-Bhagwasi watershed by application of Gypsum (50% GR) for cultivation of rice. Summer ploughing area increased from 10 ha to 47 ha in Kokriguda watershed due to its inherent advantages of weed suppression, *in-situ* conservation of pre-monsoon moisture and timely preparation of fields prior to onset of monsoon. Crop improvement was one of the major activities in all the six watershed development projects benefitting 17.6 to 136 ha area. It was achieved by introduction of HYV of crops grown with balanced use of fertilizers and new crops suitable for crop diversification. Crop demonstrations of foodgrains and vegetables were conducted for introducing the new technologies.

Non-arable lands

For safe disposal of rain water, renovation of grassed waterways (1.84 km, benefitting 7 ha), and construction of safe disposal structures (5) and diversion channel (0.2 r km, benefitting 8 ha) were undertaken in Kokriguda watershed. Similarly in Bajni watershed, a series of water outlet structures (17) in the *nallah* adjoining the recently levelled and bunded fields retarded erosivity of excess runoff and enhanced its infiltration and deep percolation thereby benefitting 123 ha of land.

Further downstream, drainage line treatment in the form of construction of checkdams and gully plugs was undertaken in Badakhera, Aganpur-Bhagwasi, Antisar and Salaiyur watersheds for protection of channels from further deepening, arresting their encroachment into agriculture fields and gully head extension. In Badakhera, 382 ha area of the ravinous watershed benefitted by this activity. Vegetative barriers of recommended grasses were transplanted along field bunds and torrent banks, across gully beds and near engineering structures for their stabilization in Aganpur-Bhagwasi (45 ha) and Badakhera (7 ha) watersheds.

Water resources development was undertaken in five watersheds through renovation/rejuvenation of existing ponds and abandoned stone quarries and repair of earthen dams. Wherever feasible, water harvesting structures like percolation/ sunken ponds and well recharge structures were constructed. Further, water conservation through HDPE lining in shallow ponds was undertaken in Salaiyur watershed. An underground pipeline system of 1,248 m length was laid in Kokriguda watershed for efficient conveyance of irrigation water.

Horticultural and agroforestry interventions were also adopted on farmers' fields/boundaries as well as in non-cultivated land for income generation and fodder production. In Aganpur-Bhagwasi, 48 ha area was brought under poplar plantation and in Salaiyur, 32 ha of marginal/degraded (LCC IV and VI) lands were planted with mango, tamarind, coconut and other fruit saplings. Some new fruit species were also introduced for diversification. Special attention was paid to fodder production by introducing HYV of grasses.

Other activities undertaken on non-arable lands to improve biomass production and minimize soil erosion were graded bunds (0.4 ha), stone bunds (2.4 ha) and contour trenches. Drainage line treatments with loose-boulder, single and double brushwood check dams (50), stone pitching (320 m) and retards (8) were all also undertaken. These measures were supported with vegetative measures like vegetative barriers (3213 r m), grass plantation (2 ha), cross barriers (3) and gully bank plantation (5 r km). Further, afforestation and pasture development works were undertaken in Antisar (37 ha) and Badakhera (38.5 ha) watersheds.

INDICATORS OF WATERSHED DEVELOPMENT

Monitoring and evaluation of various developmental interventions was carried out during the implementation phase to quantify their impact on ecology, productivity and socio-economic conditions by evolving and employing appropriate indicators/indices. The following indicators were used in pre-project and post-project scenarios to evaluate the overall impact of the watershed development programmes :

Bio-physical Indicators

Land Levelling Index

For comparing the efficiency of land levelling activity during the pre-project and post-project periods, an index called as Land Levelling Index was employed, which is the ratio of recommended land slope to the existing land slope as defined below:

$$\text{Land Levelling Index (LLI)} = \frac{\text{Recommended slope (\%)}}{\text{Existing or treated slope (\%)}} \quad (1)$$

where, existing slope refers to the individual land slope before the inception of the project and treated slope is the moderated slope resulting from land levelling activities. Higher value of LLI is a measure of better moderation in land slope. LLI can attain a maximum value of 1.0, which refers to a perfectly levelled field.

Crop Diversification Index

Crop Diversification Index (CDI) was used to assess the changes in the cropping patterns/systems due to crop improvement programmes. It is defined as :

$$\text{Crop Diversification Index (CDI)} = \sum_{i=1}^{n} P_i \log(1/P_i) \quad (2)$$

where, P_i is proportion of ith crop in comparison with total cropped area and n is the total number of crops in the watershed. The CDI can attain any value > 0 and higher value of CDI is an indicator of better crop diversification.

Cultivated Land Utilization Index

Cultivated Land Utilization Index (CLUI) is an indicator of the use of land for intercropping and multiple cropping systems. It indicates the impact of watershed interventions on changes in cultivable land area and duration of crop cultivation in pre-project and post-project periods. It is calculated by summing the products of land area planted under each crop, multiplied by actual duration of days of that crop, and dividing the sum by the cultivated land area times 365 days as given below:

$$\text{Cultivated Land Utilization Index (CLUI)} = \frac{\sum_{i=1}^{n} a_i d_i}{A \times 365} \quad (3)$$

where, n are the total number of crops; a_i is the area occupied by i^{th} crop; d_i are the days that the ith crop occupied in the a_i area; and A is cultivated land area. The CLUI can attain a maximum value of 1.0 and higher value of CLUI indicates more usage of available cultivable land area of the watershed and/or duration of crop cultivation period.

Crop Productivity Index

For assessing the overall improvement in crop productivity at the watershed level, Crop Productivity Index (CPI), which indicates the extent of crop productivity level in comparison to the normal yield of crops before and after the project, is calculated by dividing the crop yield obtained in the watershed by the yield obtained under recommended package of practices.

$$\text{Crop Productivity Index (CPI)} = \frac{1}{n} \sum_{i=1}^{n} (y_i / Y_i) \quad (4)$$

where, n is the total number of crops in the watershed, y_i is the average yield of ith crop cultivated in the watershed, and Y_i is the yield of ith crop with standard package of practices. The CPI can attain any value greater than zero in a given location. Higher value of CPI is indicative of crops' yields closer to the maximum attainable yield under standard package of practices.

Induced Watershed Eco-Index

Induced Watershed Eco-Index (IWEI) is an indicator of additional area brought under vegetative cover due to crops, pasture and grassland development, and horticultural and forestry plantations in a watershed during the project period. It is calculated as :

$$\text{Induced Watershed Eco-Index (IWEI)} = \frac{\text{Additional area vegetated during the project}}{\text{Total area of the watershed}} \quad (5)$$

The IWEI can attain a maximum value of 1.0 which indicates that whole of the watershed area has been brought under some form of vegetation.

Participatory and Socio-economic Indicators

People's Participation Index

For measuring level of participation of the community, in different phases, Peoples' Participation Index (PPI) was used. Overall PPI was computed by randomly selecting a representative sample of respondents. The responses were recorded in the pre-structured schedule through personal interviews and data were analyzed by employing appropriate statistical methods. Responses of the respondents were recorded on peoples' participation in various interventions of watershed programme and scores were assigned as 1 for "Yes"

and 0 for "No" to each statement. PPI was worked out as :

$$PPI = \frac{\text{Mean participation score (P)}}{\text{Maximum participation score}} \times 100 \qquad (6a)$$

where,

$$P = \sum_{i=1}^{N} P_i / N \qquad (6b)$$

and,

$$P_i = \sum_{j=1}^{K} (PP_j + PI_j + PM_j + PE_j) \qquad (6c)$$

In equations 6(a,b,c), PP_j are the total scores of people's participation in programme planning; PI_j are the total scores of people's participation in programme implementation; PM_j are the total scores of people's participation in programme maintenance; PE_j are the total scores of people's participation in programme's monitoring and evaluation; K are the total number of statements on which responses of the respondents were recorded; and N is the total number of respondents. Higher value of PPI is a measure of better participation in the watershed development activities. PPI can attain a maximum value of 100.0, which means participation by the whole watershed community in all the activities.

Benefit : Cost Ratio

Economic analysis of the projects was carried out separately for arable lands and non-arable lands and also for the watershed as a whole using the discounted measure of the project worth namely Benefit-Cost Ratio (BCR). It is defined as the ratio of present value of gross benefits to the present value of total costs as given below:

$$\text{Benefit-Cost Ratio (BCR)} = \sum_{t=1}^{n} \frac{B_t / (1+i)^t}{C_t / (1+i)^t} \qquad (7)$$

where, B_t are the Benefits (Rs.) at time t; C_t are the costs (Rs.) at time t; i is discount rate (%); and t is life of the project. BCR can attain any value equal to or more than zero. Projects with BCR > 1.0 are considered as economically viable and economic soundness of the project increases as the BCR value increases.

RESULTS AND DISCUSSION

Bio-physical Impacts

Land Improvement

Levelling, terracing and bunding reduced the general slope of the fields, as indicated by the Land Levelling Index.

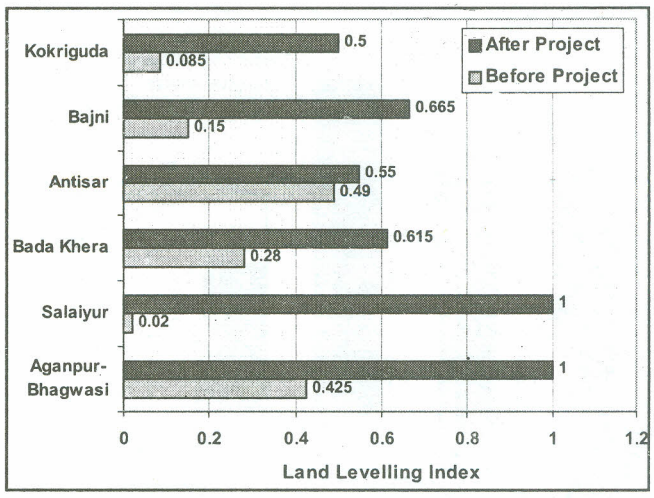


Fig. 2. *Improvement in the land levelling index value on arable lands*

Prior to these interventions, the index value ranged from 0.02 to 0.49 with weighted average of 0.37 for the six watersheds which improved to 0.5 to 1.0 with weighted average of 0.65 after the land improvement interventions (Fig. 2). This helped in uniform distribution of rainwater and availability of moisture in the soil profile which in turn improved crop yields.

Water Resource Development

Additional water storage capacity ranging from 121 to 1,584 ha-cm with an average of 474 ha-cm was created in 5 watersheds as a result of various water resource development activities (Table 5). It helped in significantly augmenting the ground water recharge which was reflected through increase in recharge rate of 20 to 73% wells located in the influence zone of the water harvesting structures by 6 to 50% in the four watersheds having open/bore wells for irrigation. The irrigated area increased by 65 to 585%, with an average of 206% giving tremendous boost to the crop yields/production and farmers' income. In Kokriguda watershed, the increase in irrigated area is attributed to underground pipeline irrigation system that diverted water from a perennial stream. During the years of low rainfall, water from the wells provided life saving irrigation to the crops thereby mitigating the effect of drought.

Table 5. *Water resource development in the IWDP watersheds*

Watershed	Water storage capacity created (ha-cm)	Wells influenced (%)	Increase in well recharge rate (%)	Irrigated area increase (%)
Aganpur-Bhagwasi	NA	NA	NA	NA
Antisar	1584	73	23	90
Bada Khera	256	20	6	65
Bajni	144	50	50	585
Kokriguda	121	NA	NA	583
Salaiyur	266	46	10 -15	84

Runoff and Soil Loss

Various mechanical and biological measures, including water resource development implemented in the watersheds, reduced runoff by 9 to 24% (Table 6). In Salaiyur watershed, pre-project runoff, which was already low, got further reduced to 1.3%. Soil loss from the watersheds was reduced drastically in the range of 0.042 to 10 t $ha^{-1}yr^{-1}$ which is well within the permissible limits. In case of Kokriguda and Badakhera watersheds, soil loss reduced by 75 to 82% from initial rate of 38 to 40 t $ha^{-1}yr^{-1}$.

Table 6. *Impact of interventions on surface runoff and soil loss in the IWDP watersheds*

Watershed	Surface runoff (%)		Soil loss (t ha^{-1} yr^{-1})	
	Pre-project	Post project	Pre-project	Post project
Aganpur-Bhagwasi	48.5	24.0	12.6	2.8
Antisar	33.0	16.0	0.405	0.042
Bada Khera	30.0	10.0	40.0	10
Bajni	25.4	16.3	12.1	8.3
Kokriguda	36.8	12.4	38.2	6.6
Salaiyur	4.5 - 7.2	1.3	1.7 - 8.9	0.5 - 1.6

Crop Diversification and Productivity

Various agronomical interventions improved the yield of traditional crops by 9% (ladyfinger, Salaiyur) to 256% (tomato, Kokriguda). The overall crop productivity of the watersheds measured in terms of the Crop Productivity Index (CPI) increased by 12 to 45% depending upon location and type of crops (Table 7). Considering five watersheds together, CPI improved from 0.50 in the pre-project period to 0.64 after the project implementation (Fig. 3). In case of Antisar watershed, the maize equivalent yield of all the crops increased by 37%.

In addition to the introduction of HYVs of traditional crops, some new crops were also introduced in the watersheds for better economic returns through crop diversification. This was well elucidated by the Crop Diversification Index (CDI), which increased by 6 to 79% in all the watersheds except Badakhera watershed where it declined by 32% (Table 7). It is ascribed to the fact that earlier the farmers were practicing subsistence type of farming systems with many crops but with better availability of resources, they switched over to lesser number of high yielding food and remunerative cash crops.

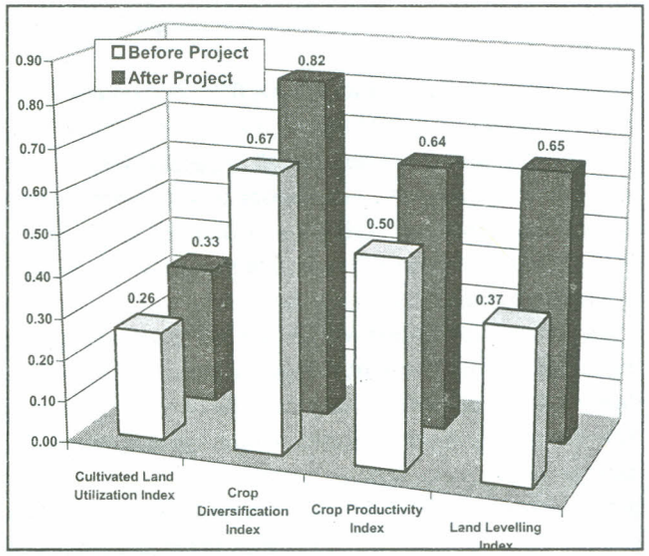


Fig. 3. *Improvement in various parameters as an indicator of overall watershed development*

However, the overall diversification of crops considering all watersheds together improved from 0.67 in the pre-project period to 0.82 after the project implementation (Fig. 3).

Cultivated Land Utilization

A cumulative effect of the interventions on agriculture improvement in the watersheds was judged through the Cultivated Land Utilization Index (CLUI), which is based on the period of the year for which the cultivated area is under different crops. The index value estimated for each of the six watersheds under study showed that land utilization improved by 2 to 81% (Table 7). Thus more and more cultivated area was brought under cultivation for longer duration as a result of agronomical interventions. The overall utilization of cultivated land of the six watersheds showed that CLUI had increased from 0.26 before the project implementation to 0.33 in the post-project period (Fig. 3).

Vegetative Cover

Vegetative measures implemented in the arable and non-arable lands in conjunction with the mechanical measures helped in soil moisture conservation, which resulted in establishment and regeneration of vegetation and provided additional green biomass

Table 7. *Crop productivity, crop diversification and cultivated land utilization indices of IWDP watersheds*

Watershed	Crop Diversification Index (CDI)		Cultivated Land Utilization Index (CLUI)		Crop Productivity Index (CPI)	
	Pre-project	Post project	Pre-project	Post project	Pre-project	Post project
Aganpur-Bhagwasi	0.55	0.76	0.16	0.29	0.49	0.68
Bajni	0.50	0.90	0.25	0.37	0.29	0.42
Badakhera	0.79	0.54	0.26	0.27	0.83	0.99
Kokriguda	0.68	0.98	0.14	0.21	0.33	0.48
Salaiyur	0.53	0.74	0.59	0.67	0.55	0.61
Antisar	0.94	0.99	0.18	0.19	NA	NA

cover to the soil in all the watersheds. The Induced Watershed Eco-Index (IWEI) varied from 0.042 to 0.28 with an average of 0.12 among the six watersheds thereby indicating that additional 12% watershed areas were rehabilitated through green biomass cover.

Participation and Capacity Building

People's Participation

The participation of the watershed community was encouraging in all the phases of watershed development programmes. The overall participation ranged from 45.3% to 79.5% with an average of 61.6% (Table 8) indicating willingness and keen involvement of the stakeholders in all the activities, which were implemented in a participatory mode as per new paradigms of rural development. Thus, the motivation of the community to get involved in the watershed development process right from the planning to maintenance stages helped in achieving good degree of success in the programme.

Table 8. *People's participation (%) in the IWDP watersheds*

Watershed	Planning	Implementation	Maintenance	Monitoring and evaluation	Overall
Aganpur-Bhagwasi	80.7	89.2	NA	68.7	79.5
Antisar	72.3	70.5	73.5	NA	72.1
Bada Khera	64	38	NA	NA	51.4
Bajni	34	62	78	56	57.5
Kokriguda	48	51	37	NA	45.3
Salaiyur	74.6	62.8	54.3	NA	63.9

Contributions by Stakeholders

An important indicator of local stakeholders' willing and active participation in the watershed development programmes is their contribution in cash and/or kind to various developmental works/activities. The contributions made by the stakeholders in the watersheds ranged from 4% (land levelling, Kokriguda) in the tribal watershed to as high as 80% (horticulture development, Badakhera) with an average of 27% depending upon location and type of activity. The cash contributions were deposited in the Watershed Development Fund (WDF) account of the respective watershed. Monetary equivalent of the contributions in kind such as labour was estimated and the same amount was withdrawn from the Watershed Project Joint Account and deposited into the WDF account. The total money deposited in respective WDF accounts ranged from Rs. 39,615/- to Rs. 4,00,188/- with an average of Rs. 1,48,636/-. This amount would be effectively utilized in maintaining the created assets during the withdrawal phase and would be further augmented by the sale of produce from the common property resources and borrowings from the credit institutions.

Income and Employment Generation through SHGs and UGs

A total of 62 SHGs were formed in the six watersheds out of which 9 were discontinued (Table 9). At Antisar, the SHGs did not function as the executive body was apprehensive of recovering of seed money and its recycling among the groups. The formation of SHGs provided additional employment to the member families ranging from 2 to 101 mandays family^{-1}year^{-1} and generated additional income ranging from Rs. 276 to Rs. 15,150/- family^{-1}year^{-1} with average seed money recovery ranging from 0 to 100%. It helped in sustaining their livelihood and confidence in watershed development programmes. None of the UGs formed was discontinued as they functioned well to carry out various land based developmental activities in the watershed under the guidance of WDT members.

Table 9. *Performance of the self help groups formed in the IWDP watersheds*

Watershed	SHGs formed	SHGs discontinued	Annual employment generated (mandays family^{-1})	Additional annual income (Rs family^{-1})	Seed money recovery (%)
Aganpur-Bhagwasi	3	0	76 - 101	2265-15150	55-72.5
Antisar	4	4	Nil	Nil	NA
Bada Khera	10	0	12-64	575-3200	0-100
Bajni	14	1	54-64	117-833	100
Kokriguda	3	0	26-74	281-385	18.5
Salaiyur	28	4	2-90	276-104407.9	100.0
Total	**62**	**9**			

Transparency Mechanism

Jointly operated Watershed Project Accounts for utilizing funds sanctioned for watershed development programmes were opened in a nearby bank branch of each watershed and 75 to 100% i.e. on an average 89% of all the expenditures were incurred on various project activities through these accounts in a transparent manner. Payments were made through cheques and records like cash books, stock registers, vouchers etc. were maintained by the Watershed Secretary subject to verification by any stakeholder, if needed.

Community Mobilization and Women Empowerment

Several General Body (GB) meetings were convened as a part of PRA exercise in each watershed by the WDT to share the knowledge and experiences of watershed communities in various sectors of watershed development programmes. It greatly helped in identifying their problems and needs, assessing the resources available with them and setting priorities for various watershed interventions. A

number of GB meetings ranging from 6 to 20 with an average of 14, were held in the watersheds in which 317 to 1,361 persons participated.

Upliftment of the women is an integral component of watershed development programmes as per new guidelines, which was achieved through their involvement in SHGs and in decision-making process as members of Watershed Committees. Each WC had 1 to 3 women as its members. Women also actively participated in the GB Meetings to highlight their problems and needs. Women participation in GB meetings ranged from 19 to 35% (on an average 28%) and as members of the Watershed Committees from 46 to 100% (on an average 69%). They also played a vital role in SHGs' meetings.

Awareness about Watershed Concept and Capacity Building

The stakeholders awareness about the watershed concept, soil and water conservation measures, PIA's role, community organization, entry point activities and linkage with other departments was assessed in each watershed following standard procedures (Fig. 4). The highest average awareness was about PIA's role (90%) followed by watershed programme concept (86%) and soil water conservation works (79%). It showed that on an average 77% stakeholders were well versed with these important components of watershed development programmes which is attributed to participatory approaches adopted in implementing various interventions. Trainings were imparted to WC members and stakeholders in income generating activities like apiculture, mushroom cultivation etc. as well as in essential requirements of participatory watershed development like account keeping etc. Number of trainings held ranged from 5 to 27 with an average of 10 and number of persons trained ranged from 62 to 386 (on an average 270) in the six watersheds. The follow-up ranged from 0 to 392% with an average value of 93%.

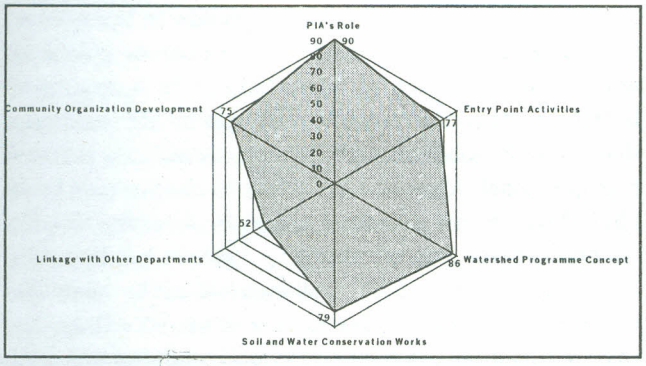


Fig. 4. *People's awareness (%) about various aspects of watershed development in the IWDP watersheds*

Socio-economic Benefits

Employment and Income Generation

A number of developmental activities were undertaken in the watersheds in a participatory mode through active involvement of the local community. No contractual system was allowed in the watersheds from outside sources in order to promote employment for the local stakeholders. Works like land improvement, water resource development and afforestation generated maximum additional casual employment. Crop improvement activities also generated sufficient permanent employment on their own fields, thus preventing their out migration for petty jobs. Overall, 7,278 (Kokriguda) to 51,461 (Antisar) additional mandays were generated during the project period in the watersheds. Thus on an average, 17,004 additional mandays of employment was generated in each watershed, which on per ha per year basis worked out to be 5 mandays. It is, therefore, established that participatory watershed management has a great potential for creating employment opportunities and ensure sustainability of the programmes through active involvement of the local community.

As evident from Fig. 5, the overall average annual income per family in the watersheds increased by 8 to 106% with an average increase of 49% due to various interventions. It is attributed to enhanced crop productivity on agricultural lands, horticulture development and income generating activities through SHGs resulting in increased employment generation. Low increase in income in Antisar watershed is attributed to continuous drought for 3 years, which hampered timely agronomical operations and resulted in low productivity. A remotely located tribal watershed at Kokriguda witnessed a quantum jump in family income, which increased

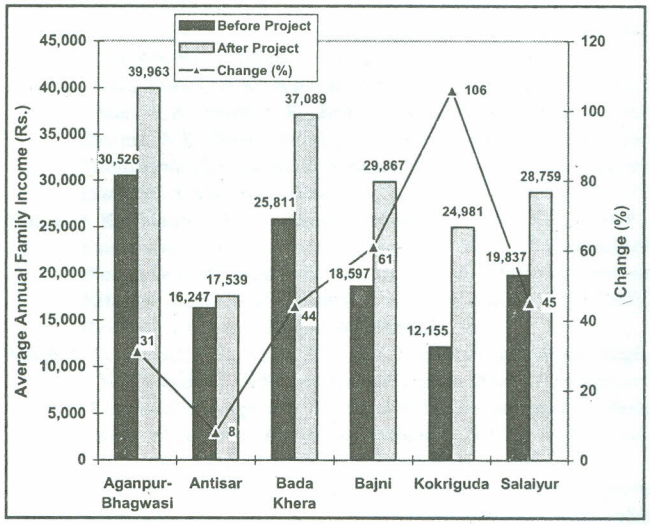


Fig. 5: *Change in annual family income of stakeholders of IWDP watersheds*

from Rs. 12,155/- to Rs. 24,981/- within a period of 5 years. It is the cumulative effect of all the biophysical and social upliftment efforts in the watershed management programmes. It amply proves that participatory planning and execution of watershed interventions can substantially improve the living standards of economically backward and disadvantaged sections of the society.

Economic Viability

The Benefit Cost Analysis of various developmental activities undertaken in the watersheds on arable and non-arable lands revealed that these projects were economically viable. The Benefit : Cost Ratio (BCR) of these projects varied from 1.14 to 1.72, with four of them having BCR > 1.5 (Table 10). In arable lands, the BCR ranged from 1.13 to 1.80 while in non-arable lands, it varied from 0.91 to 9.78 depending upon type of activity and project life. The low BCR of 0.91 in non-arable lands of Antisar watershed is ascribed to the fact that due to continuous drought for 3 years, the plantations suffered high mortality and biomass production reduced drastically.

Experiences of six watersheds developed under IWDP following the guidelines of MoRD in different agro-ecological regions have clearly established that success, sustainability and replicability of watershed development programmes largely depend upon people's participation in the planning, implementation and maintenance phases. Transparency, contributions, equity in sharing of benefits and costs, and involvement of women and other disadvantaged sections of the society are the key elements for holistic and sustainable development of rainfed areas as enshrined in the new guidelines. The benefits accrued out of various developmental activities were spectacular in all the watersheds. Several need-based interventions on arable and non-arable lands helped in efficient utilization of natural resources, which significantly improved the ecology, productivity and socio-economic conditions of the stakeholders. Overall crop productivity increased by 28% on an average while CDI improved by about 22%. Conservation measures rehabilitated the degraded lands and consequently additional 12% green cover was restored in the watersheds. With better interception of *in-situ* rainwater due to mechanical and biological measures, runoff in the watersheds reduced by 9 to 24% and soil loss was brought within permissible limits. On an average, 62% participation of the community and 27% contributions in various activities is a testimony of the success of watershed development programmes. The annual average family income increased by 49% as a result of enhanced productivity and additional mandays of employment. The projects were found to be economically viable with overall BCR of more than 1.14. It is thus concluded that participatory watershed development has tremendous potential in achieving sustainable productivity and ecological security through efficient use of natural resources.

Table 10. *Economic viability (Benefit-Cost Ratio) of the IWDP watersheds*

Watershed	Arable land	Non-arable land	Overall
Aganpur-Bhagwasi	1.64 (15,12)	2.76 (15,12)	1.72 (15,12)
Antisar	1.34 (15,10)	0.91 (15,10)	1.31 (15,10)
Bada Khera	1.80 (20,10)	1.44 (20,10)	1.54 (20,10)
Bajni	1.67 (10,12)	1.64 (10,12)	1.65 (10,12)
Kokriguda	1.65 (25,10)	1.89 (25,10)	1.69 (25,10)
Salaiyur	1.13 (20,10)	9.78 (20,10)	1.14 (20,10)

Figures in parentheses are project life and discount rate, respectively.

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